
EE 619: Mixed Signal VLSI Design

Course Code	EE 619
Course Name	Mixed Signal VLSI Design
L-T-P-C	3-0-2-4
Prerequisites	EE 512 CMOS Analog IC design
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Sample and hold and trans-linear circuits** [2 Lectures]
 - Performance of sample-and-hold circuits testing sample and holds, MOS sample-and-hold basics, examples of CMOS S/H Circuits, bipolar and BiCMOS Sample-and-Holds, Trans-linear gain Cell, trans-linear multiplier
- **Switched Capacitor circuits** [4 Lectures]
 - Basic building blocks opamps, capacitors, switches, non-overlapping clocks, Basic operation and analysis of switched capacitor circuits, resistor equivalence of a switched capacitor, parasitic- sensitive integrator, parasitic-insensitive integrators, signal-flow-graph analysis, noise in switched-capacitor circuits
 - First-Order Filters switch sharing, fully differential filters, biquad filters, low-Q biquad filter, high-Q biquad filter, Charge injection, switched-capacitor gain circuits, parallel resistor-capacitor circuit, resettable gain circuit, capacitive-reset gain circuit, correlated double-sampling techniques, other switched-capacitor circuits viz. amplitude modulator, full-wave rectifier, peak detectors, voltage-controlled oscillator, sinusoidal oscillator
 - **Comparators** [3 Lectures]
 - * Comparator specifications input offset and noise, hysteresis
 - * Opamp as a comparator input-offset voltage errors, charge-injection errors, making charge-injection signal independent, minimizing errors due to charge-injection, speed of multi-stage comparators
 - * Latched comparators, latch-mode time constant, latch offset, examples of CMOS and BiCMOS comparators, input-transistor charge trapping
- **Data converters specifications** [2 Lectures]
 - Ideal D/A converter, ideal A/D converter, quantization noise, deterministic approach, stochastic approach, signed codes, performance limitations, resolution, offset and gain error, accuracy and linearity
- **Nyquist rate digital-to-analog converters (DAC)** [4 Lectures]
 - Decoder-based converters resistor string converters, folded resistor-string converters, multiple resistor-string converters, signed outputs,
 - Binary-scaled converters binary-weighted resistor converters, reduced-resistance-ratio ladders, R-2R-based converters, charge-redistribution switched-capacitor converters, current-mode converters, glitches
 - Thermometer-code converters thermometer-code current-mode D/A converters, single-supply positive-output converters, dynamically matched current sources
 - Hybrid converters resistor-capacitor hybrid converters, segmented converters
- **Nyquist rate analog-to-digital converters (ADC)** [12 Lectures]

- Introduction to integrating converters, flash converters, issues in designing flash ADC,
- Successive-approximation converters DAC-based successive approximation, charge- redistribution A/D, resistor-capacitor hybrid, speed estimate for charge redistribution converters, error correction in successive-approximation converters, multi-bit successive-approximation
- Algorithmic (or cyclic) A/D Converter ratio-independent algorithmic converter,
- Pipelined A/D converters one-bit-per-stage pipelined converter, 1.5 bit per stage pipelined converter, pipelined converter circuits, generalized k-bit-per-stage pipelined converters
- Two-step A/D converters, two-step converter with digital error correction,
- Interpolating A/D converters, folding A/D converters, time-interleaved A/D converters
- **Oversampling ADCs** [8 Lectures]
 - Oversampling without noise shaping, quantization noise modeling, white noise assumption, oversampling advantage, the advantage of 1-bit D/A converters
 - Oversampling with noise shaping, noise-shaped delta-sigma modulator, first-order noise shaping, switched-capacitor realization of a first-order A/D converter, second-order noise shaping, noise transfer-function curves, quantization noise power of 1-bit modulators, error-feedback structure
 - System architectures system architecture of delta-sigma A/D converters, system architecture of delta-sigma D/A converters,
 - Digital decimation filters multi-stage, single stage, higher-order modulators interpolative architecture, multi-stage noise shaping (MASH) architecture, bandpass oversampling converters, Practical considerations stability, linearity of two-level converters, idle tones, dithering, opamp gain,
 - Multi-bit oversampling converters dynamic element matching, dynamically matched current source D/A converters, digital calibration A/D converter, A/D With both multi-bit and single-bit feedback
- **Phase locked loop** [6 Lectures]
 - Basic phase-locked loop architecture, voltage controlled oscillator, divider, phase detector, loop filter, the PLL in lock,
 - Linearized small-signal analysis second-order PLL model, limitations of the second-order small-signal model, PLL design example
 - Jitter and phase noise period jitter, P-cycle jitter, adjacent period jitter, other spectral representations of jitter, probability density function of jitter
 - Electronic oscillators ring oscillators, LC oscillators, phase noise of oscillators, jitter and phase noise in PLLS, input phase noise and divider phase noise, VCO phase noise, loop filter noise

Text Books

1. David A. Johns, Kenneth W. Martin, Tony Chan Carusone, **Analog Integrated Circuit Design**, 2nd Edition, Wiley, 2012.

References

1. Phillip Allen and Douglas R. Holberg., **CMOS Analog Circuit Design**, 2nd Edition, Oxford university press, 2002.
2. Willy M. C. Sansen, **Analog Design Essentials**, Springer, 2006
3. Behzad Razavi, **Design of Analog CMOS Integrated Circuits**, McGraw Hill, 2001